Data Analytics (DA)

Module-End -5

Module End Assignment: Python For Data Analysis

“Social Media Engagement Analytics”

Assignment Task Overview

Project Title: Social Media Engagement Analytics Using Python

Problem Statement

Social media platforms generate massive volumes of engagement data—likes, comments, shares, impressions, watch time, and more. Analyzing such data helps companies understand user behavior, identify trends, and improve content performance.

In this assignment, you will work with a **social media dataset** containing social media engagement metrics. The task involves **data cleaning, transformation, NumPy/Pandas operations, exploratory data analysis, visualizations**, and generating insights.

Dataset Provided: [social_media_engagement_5000.csv]

Assignment Tasks:

- ❖ **Task 1 — Data Import & Setup**
  - Import CSV using Pandas
  - Check/convert data types
  - Convert date columns to datetime
- ❖ **Task 2 — Data Cleaning**
  - ***Cleaning Missing Data***
    - Detect missing values (isnull(), isna())
    - Handle using: dropna(), fillna(), median/mode, forward/backward-fill
  - ***Duplicate Handling-*** Identify & remove duplicates
  - ***Data Formatting***
    - Fix incorrect data types
    - Standardize categories (e.g., gender labels)
    - Correct unrealistic values in likes, comments, shares

- ***Feature Cleaning***
    - Extract hashtag count
    - Clean sentiment labels

❖ **Task 3 — Data Exploration using Pandas**

Perform the following:

- View dataset structure using head(), tail(), shape, and columns.
- Check data types and info with info() and dtypes.
- Generate summary statistics using describe().
- Analyze categorical distributions using value_counts(), unique(), and nunique().
- Create a correlation matrix for numeric fields.
- Use groupby() to summarize metrics (e.g., avg likes by post type, impressions by country).

❖ **Task 4 — Data Wrangling**

- Use **merge**, **concat**, or **join** if combining DataFrames.
- Create new fields such as *engagement_score, log-transformed metrics* (optional), and hashtag count.
- Perform **groupby** summaries by **post_type**, **country**, and **sentiment**.

❖ **Task 5 — Statistical Analysis**

Compute descriptive stats for "**likes, comments, shares, watch_time, engagement_rate, followers"** columns:

- Mean, median, mode
- Standard deviation, variance
- Percentiles
- (Optional) Skewness and kurtosis

❖ Task 6 — Data Visualization (Min. 8 Plots Required)

- **Matplotlib**
    - **Scatter:** *likes vs impressions*
    - **Line:** *daily engagement trend*
    - **Bar:** *posts by category*
    - **Pie:** *gender distribution*
    - **Histogram:** *age*
    - **Box:** *engagement rate*
- **Seaborn**
    - **Count plot:** *post type*
    - **Bar plot:** *avg likes by category*
    - **Violin:** *followers vs sentiment*
    - **Pair plot:** *numeric features*
    - **Heatmap:** *correlation matrix*
    - **Swarm plot:** *engagement vs device*
- **Plotly (Interactive)**
    - *Interactive line chart/bar chart/bubble/scatter chart*

❖ Final Insights should include the following analysis:

- **Content Performance**
    - *Which post types have the highest engagement?*
    - *Best-performing content category?*
    - *Which countries have the highest average engagement rate?*
- **User Trends**
    - *How age affects engagement*
    - *Performance difference for verified accounts*
- **Behavioral Insights**
    - *Best time of day for impressions*
    - *Device type impact on watch time*
- **Sentiment Analysis**
    - *Which sentiment performs best*
    - *Behavior of negative/neutral sentiment posts*

Deliverables

- Complete the assignment in Google Colab or Jupyter Notebook.
- The Python file must include the following: *Code, Outputs, Visualizations, and Explanations for each code, wherever possible.*
- For Google Colab, share the notebook link with "View" access. If using Jupyter, upload the notebook to Google Drive and ensure "View" access is enabled.
- Clear one-page summary with findings supported by analysis

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Expected Level
Data Loading & Setup	Python + Pandas	CSV loading, dtype checks, and datetime conversion	Beginner
Data Cleaning & Transformation	Data Cleaning	Missing values, duplicates, formatting, corrections	Intermediate
Exploration & Wrangling	Data Handling	EDA, slicing, indexing, groupby, new features	Intermediate
Statistical Analysis	Quantitative Analysis	Summary stats, distributions	Intermediate
Data Visualization	Visualization	Matplotlib, Seaborn, Plotly charts	Intermediate
Insights & Reporting	Communication	Interpret trends and patterns	Beginner

Rubric & Mark Allocation (Total 25 Marks)

Skill/Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Credit
Data Loading & Setup	**3**	Dataset imported correctly, dtypes verified, datetime converted.	Minor issues in import, dtype handling, or incomplete setup	**1–2**
Data Cleaning & Transformation	**5**	Missing values handled, duplicates resolved, formatting fixed, categories standardized, sentiment & hashtags cleaned	Some cleaning steps were done, but incomplete or inconsistent	**3–4**
Exploration & Wrangling	**5**	Complete EDA: head/tail/info/describe, correlation matrix, slicing/indexing, new features, and valid groupby summaries	EDA is correct, but wrangling is incomplete OR missing some derived fields	**2–4**
Statistical Analysis	**4**	Accurate descriptive stats for likes, comments, shares, followers, watch_time, engagement rate	Most statistics are correct, but incomplete or missing fields	**2–3**
Data Visualization	**6**	≥ 8 plots across Matplotlib, Seaborn, Plotly; labeled, clear, and meaningful	Missing some plots, poor labeling, or incomplete library coverage	**2–5**
Reporting & Insights	**2**	Clear summary explaining trends, engagement insights, and key patterns	Summary is present but shallow, unclear, or unsupported	**1**

Skill Taxonomy Structure

- **Data Loading & Management**
  - Load CSV with Pandas
  - Convert data types & datetime
  - Create/use NumPy arrays
  - Organize columns and indexes
- **Data Cleaning**
  - Handle missing values & duplicates
  - Standardize text and categories
  - Fix invalid numeric values
- **Analysis**
  - Compute descriptive statistics
  - Create new metrics (engagement rate, scores)
- **Data Summarization**
  - Sort, filter, group data
  - Compute aggregates with groupby()
  - Find uniques, distributions, and correlations
- **Data Visualization**
  - Matplotlib & Seaborn plots
  - Plotly interactive charts
  - Proper labels, titles, formatting
- **Reporting & Insights**
  - Summarize key findings
  - Explain trends and patterns
  - Link metrics to meaningful insights

Evaluation Guidelines

- Review the complete code, including code, outputs, and explanations.
- For each section, match student work to the specific skill and sub-skill area in the taxonomy.
- Assign full or partial marks per the rubric, using suggested partial mark ranges for "partially complete" work.
- For ambiguous or incomplete submissions, note what's missing and correspondingly reduce credit.
- Comments must explain strengths and improvement points for each section.
- Objective assessment: focus on function, correctness, and clarity, not on visual design beyond the stated requirements.

Achievement Criteria & Passing Rules

- ➢ At least partial (not zero) credit must be received in every section—no component can be left unattempted.
- ➢ Students must fully achieve at least two major skills**:**
  - **Data Cleaning & Preparation, Data Wrangling & Transformation, Exploratory & Statistical Analysis, Data Visualization**
- ➢ Feedback and a detailed scorecard must be given for all evaluated assignments.